

SCIODYSSEY

DISCOVERY AMPLIFIED

AUTONOMOUS ML RESEARCH / TECHJAM 2026

SciOdyssey.

A persistent research world
for fresh scientific trajectories.

PRESERVE EVIDENCE. REOPEN JUDGMENT.

Research agents can execute. Discovery still breaks.

Long-horizon research is not one tool call. It is a changing environment with expensive context and uncertain recovery.



Too much context

Understanding a file costs far more tokens than changing it. The agent pays that reading cost again.

CONTEXT COST



One trajectory

A long-lived context carries yesterday's momentum into today's search.

ANCHORING



Closed workflow

Predefined tools and recovery paths stop when the environment produces something unexpected.

OPEN-WORLD GAP

Research is an **open-world** task. We need an **open-world coding agent**.

When the next capability is unknown, the system must be able to inspect the environment, extend it, and continue.

01

Open action space

⌘ Shell □ Files
🌐 Browser 📁 Git

USE WHAT EXISTS

02

Self-recovery

! Read errors 📄 Inspect docs
</> Inspect source + Install

FIX WHAT BREAKS

03

Runtime extension

🔗 MCP ⌘ Skills
> _ CLI + Packages

ADD WHAT IS MISSING

WORKFLOW EXAMPLE

LangGraph makes a known path explicit as nodes and edges.

Open-world research must leave the graph when a capability is missing.

An AI collaborator for scientific discovery.

One persistent world. Two scientific roles. A runtime that enforces mechanics.

RESEARCH WORLD / ON DISK

Evidence	Metrics, failures, original reports
Curated priors	Current understanding, open to revision
Code + State	Recoverable implementation and provenance

SCIENTIFIC JUDGMENT



Scientist

Hypothesize, code, train, evaluate, interpret.

REPORTS EVIDENCE

PERSISTENCE AUTHORITY



META

Audit claims, scope conclusions, preserve memory.

RUNTIME

Deterministic mechanics.

- Isolation
- Cancellation
- Runner + evaluator
- Logging and artifacts

The system protects the world.
The Scientist chooses the science.

Keep the world. Reset the researcher.

The persistence boundary separates what should survive from what should be allowed to start again.

01

PERSISTS

Research world

- ✓ Evidence and experiment ledger
- ✓ Failures and original reports
- ✓ Retained implementation + State
- ✓ Research preferences and constraints

CONTEXT RESET



Inherit experience, not prior reasoning.

02

RESTARTS

Fresh Scientist

- New scientific judgment
- Independent research trajectory
- New hypothesis selection
- Recoverable handoff from shared memory

From task contract to retained evidence.

One bounded autonomous loop turns a plain research brief into checked artifacts and an auditable record.

`task.md`

What to solve

Objectives · constraints · evaluation

Rank short videos for each user and interaction context.

`PERSONAL.md`

How to solve it

Preferences · research style · priorities

Run autonomously. Preserve evidence. Keep credentials safe.

ONE AUTONOMOUS CYCLE

E001 → E013

01 Define

Read the contract and inspect the environment

02 Run

Code, train, evaluate, and log the result

03 Recover

Read failures and repair the experiment

04 Retain

Keep evidence, code, and a scoped conclusion

RECOVERY OBSERVED

Failure became evidence.

NumPy float32 values broke JSON serialization. The Scientist converted the scalars, reran, and retained the measurements.

ERROR REPAIR RERUN RETAIN

Verified result. Auditable run.

Public validation shows the retained recipe and the record behind it.

PRIMARY / PUBLIC VALIDATION

0.6059363

+0.0043363

absolute improvement over Official FM

46 categorical features
eight-seed FM ensemble · NumPy / CPU

Primary score

0.6000 — 0.6060

0.6016000



Official FM

0.6059363



SciOdyssey

Primary = mean(GAUC, nDCG@5) · public validation only

Four things we learned on the journey.

The run demonstrates a useful system boundary, while leaving important causal questions open.

01

Fresh trajectories interrupt anchoring

Externalized evidence survives. Unfinished reasoning does not have to.

02

Context remains the bottleneck

Understanding files and surrounding assumptions can dominate the edit itself.

03

Recovery is part of science

A failed run can become a retained repair and a better-scoped conclusion.

04

Breadth needs evidence

Parallel search is a capability. Its causal contribution still needs ablation.

OBSERVED, NOT PROVEN

One task establishes a working boundary, not generality. META, State, and reset effects are not isolated.

THE RESEARCH WORLD CONTINUES

One research world. Many possible trajectories.

Preserve evidence. Reopen judgment. Let
the next Scientist take the work further.

github.com/zc6600/research-agent-kuairand ↗